

SOFTWARE AI vs PHYSICAL AI

Aspect	AI	Physical AI (Robotics/Embodied AI)
Nature of system	Purely digital, runs on apps, cloud, or devices	Physical systems operating in real-world environments
Scaling method	Fast digital deployment (internet-based distribution)	Slower due to manufacturing, installation, and maintenance
Main scaling drivers	Compute power, algorithms, data availability	Cost of hardware + adoption in real environments
User adoption factor	Convenience, performance, features	Trust, safety, and comfort with physical presence
Environment of use	Screens, servers, mobile devices	Homes, hospitals, workplaces, shared physical spaces
Data collection	Mostly digital behaviour (clicks, searches, usage patterns)	Continuous real-world sensing (cameras, audio, sensors)
Privacy concerns	Data tracking within apps and platforms	Much higher due to constant monitoring in private spaces
Security risks	Data breaches, account misuse	Physical + digital risks (hacking, surveillance, misuse of sensors)
Data ownership clarity	Often defined via terms of service	Still evolving; unclear standards for robots in homes
Adoption speed	Rapid global adoption	Slower due to regulation, trust, and safety concerns
Key barrier	Competition and compute costs	Safety, privacy, and ethical governance in real environments

at a viable cost. While affordability continues to improve, the ability to deliver practical value has expanded significantly, accelerating adoption across industries.

As a result, robotics is expanding beyond traditional industrial applications into areas such as agriculture, healthcare, cooking automation, hospitality, logistics, and delivery.

The new software layer transforming robotics

Modern robotics is no longer just about mechanical design and control systems. A new software layer—often referred to as VLA and, more broadly, VLM-based systems—is increasingly being used as the intelligence layer within robots, forming a key component of what is now known as Physical AI.

A robot is essentially a combination of hardware and software. From the hardware perspective, this includes structures, motors, wiring, sensors, and electronic systems. Traditionally, robotic software relied on predefined logic and machine learning models. As a result, robots typically performed well only on

tasks for which they had been specifically trained and often struggled to adapt to unfamiliar situations in the real world.

This is where VLA models come into play. These models aim to help robots understand the world more like humans by combining visual perception, language understanding, and action.

Rather than relying solely on explicit programming, robots can learn from a wide range of sources, particularly videos of people performing tasks in real-world environments. This helps them develop a broader understanding of their surroundings and improve their ability to adapt to diverse situations. VLA and other world models are being developed to enable robots to interpret their environments, determine appropriate actions, and learn from large and diverse datasets.

Global participants developing and deploying such systems include Boston Dynamics, which is known for advanced mobile robots; Tesla, which is developing humanoid robots; Agility Robotics, which specialises in humanoid robots for

warehouse applications; and Unitree Robotics, which is developing agile quadruped and humanoid robotic systems. The majority of systems focused on VLA and world models remain in the research phase or early stages of deployment, with an emphasis on improving robotic autonomy and data-driven learning.

Similar concepts are also being explored by Indian robotics companies. For example, Addverb Technologies is working on humanoid systems and warehouse automation, while iHub Robotics and other startups are developing technologies that enable robots to learn autonomously. These companies are beginning to utilise VLA-like AI within their technology stacks.

Physical AI aims to create robots that can perceive dynamic environments, make informed decisions independently, and adapt their actions in real time.

A longer-term goal is the development of general-purpose robots capable of performing multiple roles across different environments with minimal instruction.

The hidden chaos behind physical AI

One of the biggest challenges is data collection and transformation. In most cases, data is well-defined and easy to collect. Robotics and physical AI, however, require large amounts of unstructured data. This includes videos of people, data from robot-mounted sensors, and information about the surrounding environment. The biggest difficulty is not collecting the data; it is transforming it into a format from which robots can learn skills and knowledge and apply them across multiple tasks.

Companies working in this area collaborate with industries to gather operational data, particularly from manufacturing and domestic environments. This raises significant privacy, security, and video-surveillance concerns. Companies may also be reluctant to share internal